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Timing of the North-South China collision from new middle to late Mesozoic paleomagnetic data from the North China Block

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Abstract. We collected paleomagnetic samples from 43 sites of Middle and Upper Jurassic and Cretaceous rocks along a 250 km wide east-west trending belt located north of the suture between the North and South China Blocks. In each case, laboratory treatment isolates dual polarity, antipodal magnetizations that are demonstrably pre-folding. The paleomagnetic poles show very good coherence with coeval poles from other localities from the North China Block (NCB), suggesting the apparent polar wander path (APWP) from the NCB is not distorted by post-Middle Jurassic local vertical axis block rotations. A comparison of the NCB APWP with that from the South China Block suggests that collision continued after the Early Jurassic and ended no later than the Late Jurassic. A cusp in the NCB APWP coincides with a marked angular unconformity that separates Middle Jurassic from Upper Jurassic and younger sediments. Together, these observations indicate that the scissor-like collision and associated deformation ceased at the Middle to Late Jurassic boundary (~159 Ma). The time of rapid (1°/Ma) relative angular velocity between the two plates (225 to 190 Ma) coincides with a peak in U-Pb and Ar-Ar dates obtained from metamorphic rocks in the suture. The combined Late Jurassic and Cretaceous paleomagnetic data from both blocks suggest that the united China plate traveled 2000 km north relative to Eurasia during the Late Jurassic before docking in the Cretaceous. This younger collision event is also marked by a cusp on the Cretaceous part of the Eurasian APWP.

Introduction

The eastern half of China contains two major tectonic blocks, the North China Block (NCB) and the South China Block (SCB), that are separated by the east trending Qinling (Dabie) suture (Figure 1). The suture is characterized by high-pressure, coesite and diamond bearing eclogites and a complex history of arc accretion [e.g., *Wang et al.*, 1989; *Xu et al.*, 1992a]. The eastern part of the Qinling belt is truncated by the north trending Tancheng-Lujiang (Tan-Lu) fault system (TLF on Figure 1). About 500 km north of the Qinling belt, another coesite bearing, high-pressure metamorphic complex crops out in Shandong and Jiangsu provinces, east of the Tan-Lu fault [*Hirajima et al.*, 1990; *Enami and Zang*, 1990]. Similarities of metamorphic mineral assemblages and Precambrian stratigraphy suggest both metamorphic complexes were once part of the same belt that was later displaced by sinistral shear along the Tan-Lu fault [*Xu et al.*, 1987; *Xu et al.*, 1992b; *Okay and Sengör*, 1992; *Xu*, 1993; *Li*, 1994]. East of the Tan-Lu fault and south of the Shandong-Jiangsu (Sulu) metamorphic belt there are places where the Paleozoic stratigraphic sequence is typical of the SCB [e.g., *Zhang et al.*, 1984].

Early estimates for the age of amalgamation between the two blocks vary from Precambrian to Triassic with geologic evidence (age of metamorphism, etc.) yielding older ages [*Huang*, 1977; *Mattauer et al.*, 1985] and the younger ones stemming from paleomagnetic studies [*McElhinny et al.*, 1981]. Similar Carboniferous and Permian fauna and flora assemblages suggest the NCB and SCB were not widely separated in the late Paleozoic [*Laveine et al.*, 1987]. A tectonic reconstruction of the collision based on paleomagnetic data suggests that the SCB and NCB first collided near the eastern part of the suture in the Late Permian, then progressively scissored together as the SCB rotated clockwise with respect to the NCB [*Zhao and Coe*, 1987, 1989]. The progressive westward transition from marine to continental sedimentation along the suture supports this model [*Yin and Nie*, 1993]. Thus initial contact of the two plate margins likely occurred in the eastern part of the suture in the latest Paleozoic.

The U-Pb and Ar-Ar isotopic dates of metamorphic rocks in the suture range from the Precambrian to the Cretaceous. Interpretation of the dates in recent studies agree that the older ones are related to various tectonic events such as island arc collisions, while the younger, especially those between 235 and 170 Ma, are related to a more extreme collision phase between the two major plates [*Reischmann et al.*, 1990; *Mattauer et al.*, 1991; *Ames et al.*, 1993, 1996; *Eide et al.*, 1994; *Hacker and Wang*, 1995].

Age estimates for the end of collision and related deformation are ambiguous. Although some Ar-Ar dates of metamorphic rocks are early to middle Mesozoic in age, equally important are those with Cretaceous ages [e.g., *Eide et al.*, 1994]. The younger dates could be related to exhumation of the metamorphic terrains or due to argon loss during heating from igneous intrusion and volcanism which occurred along the length of the suture in the Late Jurassic to Early

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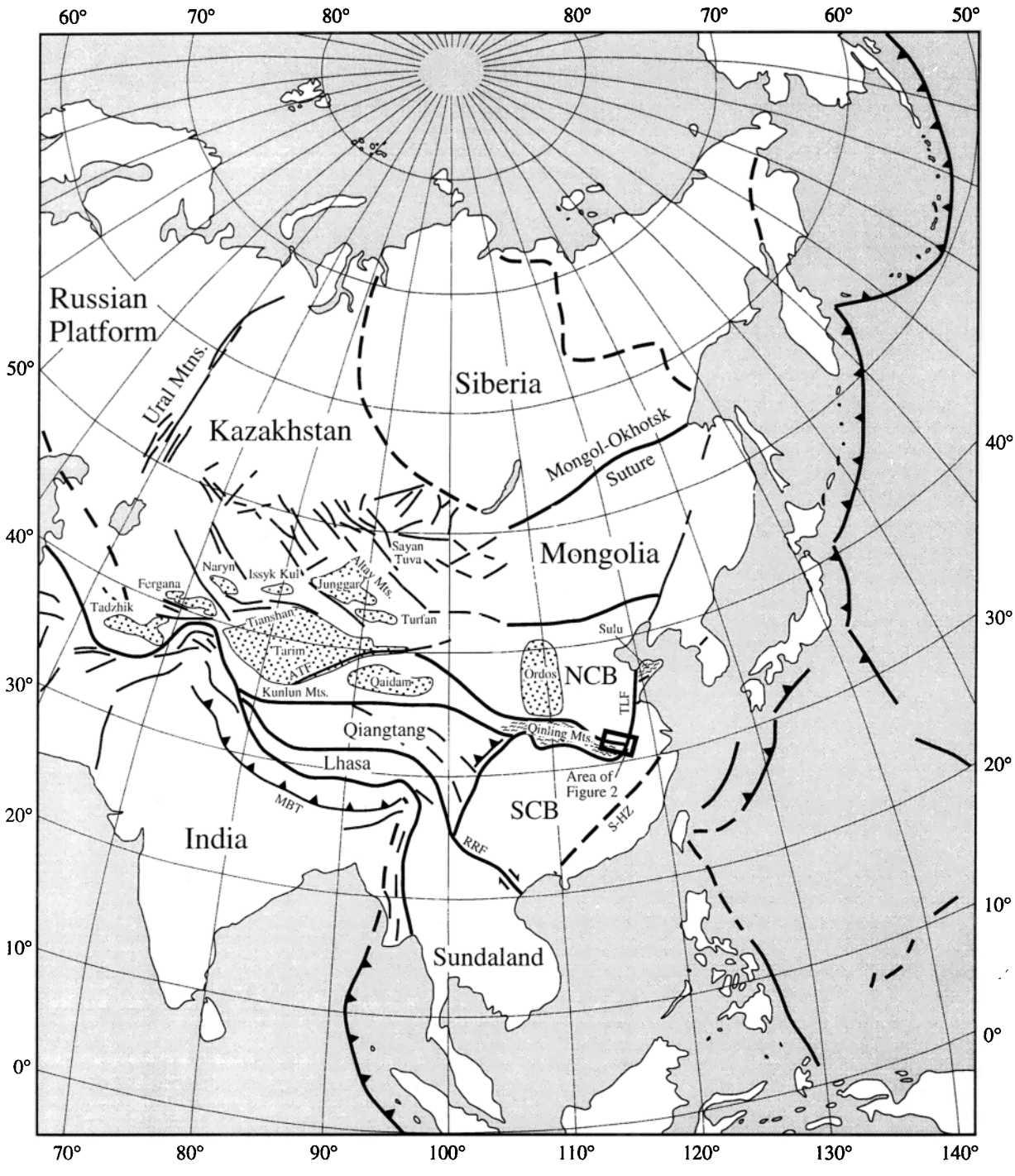


Figure 1. Regional tectonic map of Asia. Some basins and mountain chains are shown on the figure. Abbreviations are MBT, Main Boundary Thrust; RRF, Red River fault; ATF, Altyn-Tagh fault; TLF, Tancheng-Lujiang fault; S-HZ, Shi-Hang Zone; NCB, North China Block; SCB, South China Block. Speckled shading represents the location of selected basins.

Cretaceous [*Academia Sinica*, 1976]. Precise paleomagnetic estimates regarding the end of collision are thwarted because only a few post-Middle Triassic to pre-Cretaceous data exist from both North and South China. For example, the Middle and Late Triassic and Early to Middle Jurassic poles from the NCB are derived from singular studies that come from the Ordos basin [*Yang et al.*, 1991, 1992]. Because the strata there are flat-lying, no fold test can be performed, and the interpretation that primary magnetizations exist in the rocks relies on the presence of dual polarities. It is also unclear whether the data from Ordos are representative of the entire NCB. Based on comparisons of the apparent polar wander paths (APWP) from the North and South China blocks, *Huang and Opdyke* [1991] proposed that the final suturing occurred by the Late Jurassic, whereas *Yang et al.* [1992] concluded that their accretion ceased in the Middle Jurassic. In a paleomagnetic study of Lower Jurassic rocks from the SCB, *Gilder et al.* [1993] concluded that 20° to 40° of relative

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rotation between the NCB and SCB occurred after the Early Jurassic.

In this study we present Middle and Late Jurassic and Cretaceous paleomagnetic data from the NCB. These data are among the first from the NCB to yield positive fold tests. The poles are indistinguishable from coeval reference poles from the Ordos basin, which further substantiates the reliability of the paleomagnetic data from there and supports the notion that the NCB behaved as a rigid block. Together with the deformation patterns we observed in the field, the paleomagnetic data constrain the history of the NCB-SCB collision.

Geology and Sampling

We sampled middle to upper Mesozoic volcanics and red bed, lacustrine, and volcanoclastic sedimentary rocks along a 250 km east-west trending belt just north of, and parallel to, the high-pressure metamorphic rocks that comprise the Qinling-Dabie suture in Anhui and Henan provinces (Figure 2). Normally, a site represents 8 to 10 samples obtained from a suite of sedimentary layers, spanning several meters, to tens of meters in thickness; samples from sites containing volcanic rocks generally come from single flows or cooling units. The strata that we sampled contain no marine marker taxa. Our age data comes from the provincial geologic compilations. To our knowledge, these are the best references available, representing several decades of cumulative work. We know that, for whatever reason, not all the data available to the geologic surveys are published in the compilations. However, they are the state of the art for current knowledge on formation ages. The age data are used here in the same way and use the same logic as most of the paleomagnetic studies carried

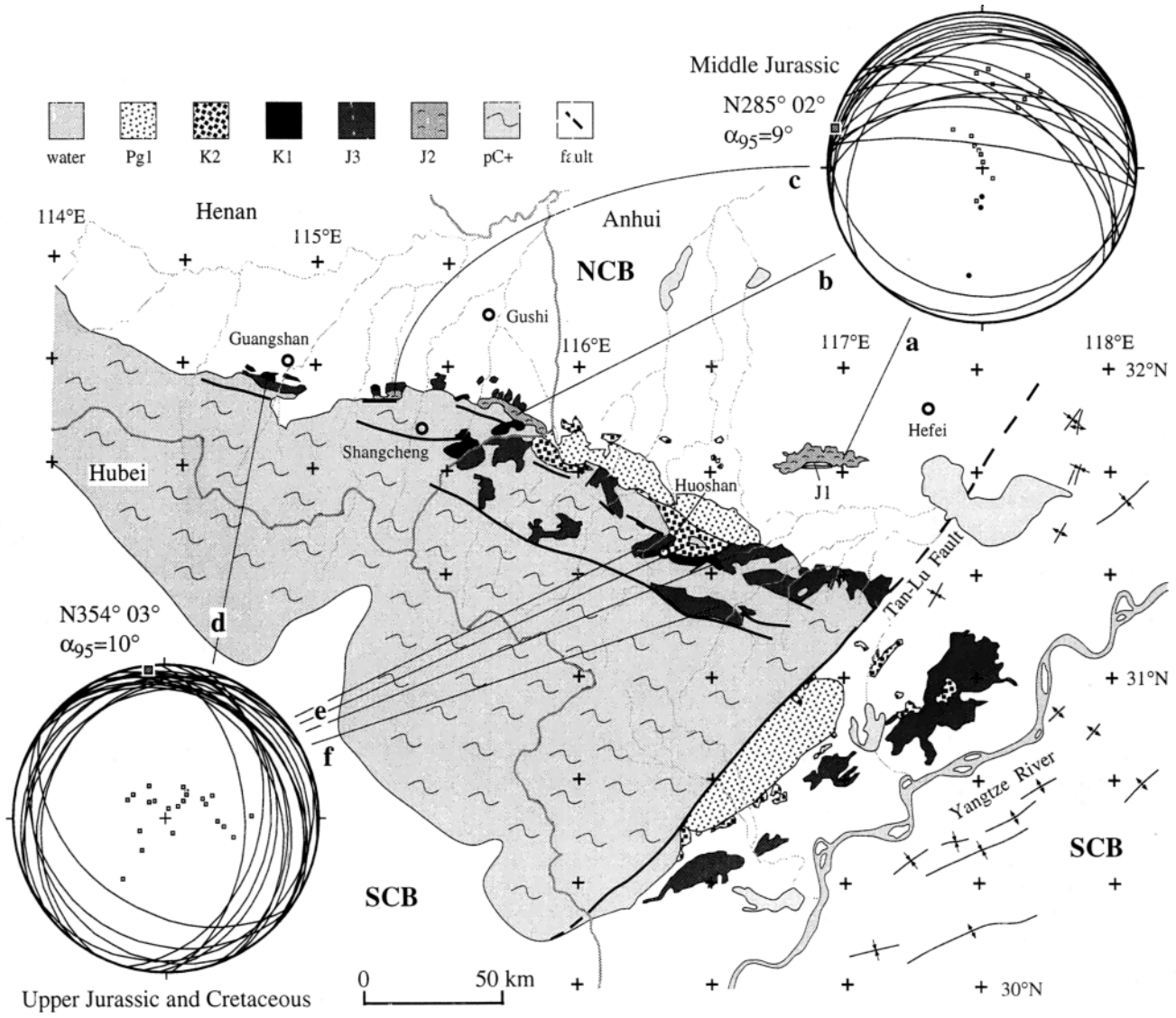


Figure 2. Geology of the eastern part of the Qinling suture between the North and South China blocks (from *AHBGMR* [1987], *HNBGMR* [1989], and field notes). Abbreviations are pC+, mostly Precambrian and younger metamorphic and igneous rocks; J, Jurassic; K, Cretaceous; Pg, Paleogene; 1, Early; 2, Middle; 3, Late (2, Late for the Cretaceous). Poles to bedding and bedding planes measured for this study are shown in the equal-area stereonets. Letters near the stereonets correspond with those in Table 2.

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out in China. We accepted the ages from the provincial geologic maps and interpreted the data/results accordingly.

In Henan Province we collected five sites of Lower Cretaceous air fall tuff (Chenpeng Formation) (Figure 2, locality d). Four of the Cretaceous sites consist of red air fall tuff interlayered with thinly bedded sandstones, the latter of which were used to orient the flows. One flow of purple-red air fall tuff that overlies the red air fall tuffs was also sampled. The Chenpeng Formation contains plant fossils of *Piceoxylon manchuricum Sze.* and estherias: *Eosestheria sp.*, *E. elongata* (Kobayashi et Kusumi), *E. aff. middendorffii* (Jones). and *Bairdestheria sp.* and unconformably overlies the Upper Jurassic Duanji Formation [*Henan Bureau of Geology and Mineral Resources* (HNBGMR), 1989, pp. 250-251]. A flow within the 5200 m thick Chenpeng Formation was isotopically dated (method unspecified) at 96.5 to 112 Ma [*HNBGMR*, 1989, p. 251].

We collected nine sites of Middle Jurassic red medium- to fine-grained sandstones and siltstones (Zhuji Formation) at two locations in Henan Province: one in Shangcheng county and the other in Gushi county (Figure 2, localities b and c). The outcrops in both places are considered Middle Jurassic because they are (1) thought to correlate with the fossil-rich Middle Jurassic Ma'ao Formation and (2) unconformably overlain by volcanoclastic sediments (Duanji Formation) whose ages are constrained by spore and pollen fossils as Upper Jurassic [*HNBGMR*, 1989, p. 237]. The red beds we sampled in Gushi county are intruded by 142 Ma felsic dikes [*HNBGMR*, 1989, p. 243]. At the Shangcheng locality, Proterozoic schists overthrust the Zhuji red beds, the latter of which contain large schistose clasts near the contact. The south dipping beds that we sampled there were mapped as overturned by previous workers (Wang Daoxian, field notes, 1990).

In Anhui Province (Figure 2, locality a) we collected 10 sites of Middle Jurassic Zhougongshan Formation red siltstones and fine-grained sandstones, 40 km to the southwest of Hefei city. These red beds disconformably overlie Lower Jurassic Fanghushan Formation white-, yellow-, and gray-colored sandstones and siltstones that are locally interbedded with coal seams. According to the *Anhui Bureau of Geology and Mineral Resources* (AHBGMR) [1987, p. 151], the Zhougongshan sediments are coeval with the nearby Middle Jurassic Luoling Formation [*AHBGMR*, 1987, p. 162]; the latter of which contains plant fossils, *Radicites sp.*, *Elatoclatus sp.*, and *Cladophlebis sp.*; gastropods, *Amnicola cf. Zhejiangens and Aphanotylus sp.*; ostracods, *Darwinela impudica, D. changxinensis,* and *D. cf. oblonga*; and bivalves, *Lampurotula (Eol.) cremeri, L. (Eol.) guangyuanensis, Psiluno sp., P. ovalis, P. chaoi, P. giganteus, P. aff. suni, P. cf. sinensis, Cuneopsis sichuanensis, C. sp., Pseudocardinia sichuanensis, P. kweichouensis, P. minuta, Margaritifera sp., Sibireconcha sichuanensis,* and *Tutuella rotunda* [*AHBGMR*, 1987, pp. 153-154].

In and around Huoshan county, we sampled 10 sites of Maotanchang and Heishidu Formation red beds, andesites, and volcanoclastic sedimentary rocks. The Upper Jurassic age assigned to these formations is constrained by abundant plant, bivalve, gastropod, and estheria fossils [*AHBGMR*, 1987, pp. 151-152]. Also in and around Huoshan county, we sampled two sites of Lower Cretaceous Shibata Formation red beds and four sites of Upper Cretaceous Xiafuqiao Formation red beds. Remains of plants, bivalves, estherias, and ostracods found within the red beds constrain their ages [*AHBGMR*, 1987, p. 167]. In Yecheng County we collected three sites of Heishidu Formation black shales (Figure 2, locality f). These rocks are interesting because we have not observed upper Mesozoic, lacustrine, sulfide and organic rich, relatively deep water, rhythmically bedded deposits like the Heishidu shales elsewhere in south or east China.

Published stratigraphic sections in Huoshan county, where some of our samples were collected (Figure 2), show an angular unconformity between Middle Jurassic and Upper Cretaceous rocks [*AHBGMR*, 1987, p. 168]. Other sections in nearby counties identify angular unconformities between Middle and Upper Jurassic strata, and Middle Triassic and Upper Jurassic strata, respectively [*AHBGMR*, 1987, p. 155]; in the Hefei basin (Figure 2, locality a) Middle Jurassic strata conformably overlie Lower Jurassic sediments, elsewhere, Lower Jurassic rocks conformably overlie Upper Triassic rocks [*AHBGMR*, 1987, p. 153]. Altogether, these sections suggest that an angular unconformity separates Middle from Upper Jurassic strata.

Included in Figure 2 are stereonets of the poles to bedding and bedding planes of the strata that we sampled for this study. The data are separated into two groups: Middle Jurassic and Late Jurassic through Cretaceous. The bedding orientations of the Middle Jurassic strata define an axial plane that parallels the east-west trending Qinling suture (N285° 02°, α_{95}=9°), whereas the Late Jurassic through Cretaceous strata have a fold axis that trends north-south (N354° 03°, α_{95}=10°), orthogonal to the trend of the Middle Jurassic fold axis. These fold axis trends are significantly different at 95% confidence limits. Their difference in orientation further indicates that a regional tectonic event separates Middle Jurassic from younger sediments. Folding of the Middle Jurassic red beds thus likely occurred soon after deposition. Upper Jurassic and Cretaceous rocks were probably folded in the Cenozoic; beds of these ages that we sampled rarely have dips exceeding 30° (Table 1).

Because (1) the axial plane of the Middle Jurassic folds parallels the Qinling-Dabie suture and (2) the axial plane defined by the folds of the younger sediments is orthogonal to the suture, it would seem that north-south directed compression acted in the Middle Jurassic and ceased in the Late Jurassic. Field and aerial mapping in the Qinling zone indicate that south China began extruding east with respect to north China in the Cenozoic and continues its eastward course relative to the NCB today [*Peltzer et al.*, 1985; *Zhang et al.*, 1995]. Thus Cenozoic folding may be linked to the India-Asia collision when the Qinling suture was reactivated as a left-lateral strike-slip fault in order to accommodate the extrusion of the SCB

Paleomagnetic Results

We collected the samples with a portable drill, oriented them with a magnetic compass, and took Sun compass bearings to correct for potential declination anomalies. Thermal demagnetization and magnetic measurements of the samples collected in 1995 were carried out in a shielded room at the Institut de Physique du Globe de Paris using a custom-built oven, alternating field demagnetizer, and three-axis CTF cryogenic magnetometer. A pilot study of four sites (C371-C374) collected in 1993 were thermally demagnetized at the University of California at Santa Cruz and measured using a

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Table 1. High-Temperature Component Paleomagnetic Data From This Study

Site	Lith.	Loc.	no/No	N/R	*Dg*	*Ig*	*Ds*	*Is*	*k*	α_{95}	DD	Dip
						Middle Jurassic						
C371	red beds	a	5/10	5/0	46.0	71.2	20.6	29.5	38.5	12.5	7.2	45.0
C372	red beds	a	7/8	0/7	71.4	76.1	18.6	30.8	55.0	8.2	3.6	53.0
C373	red beds	a	8/9	8/0	17.0	43.1	13.4	32.1	72.7	6.5	341.9	12.0
C374	red beds	a	9/9	2/4	22.3	36.4	21.9	33.7	92.3	5.5	11.3	2.8
C506	red beds	c	5/7	3/2	195.8	-18.4	353.6	48.9	48.6	11.1	6.7	149.0
C507	red beds	c	9/9	7/0	169.1	43.7	10.4	30.3	9.8	17.4	0.7	105.0
C508	red beds	c	9/9	9/0	176.9	60.1	14.8	41.2	86.2	5.6	7.7	78.0
C509	red beds	c	7/7	1/6	358.8	65.3	342.3	42.9	67.6	7.4	322.7	25.0
C510	red beds	c	8/8	8/0	201.9	26.1	338.2	39.0	51.4	7.8	1.7	111.0
C513	red beds	b	9/9	0/6	343.4	81.3	20.1	28.0	119.3	4.8	26.9	55.5
C514	red beds	b	6/6	0/6	344.0	60.4	6.9	29.6	96.8	6.8	31.8	37.0
C534	red beds	a	9/9	9/0	28.7	43.8	20.6	30.3	78.6	5.9	341.5	17.0
C535	red beds	a	8/8	8/0	38.5	33.1	34.6	27.0	67.4	6.8	349.2	9.0
C536	red beds	a	8/8	0/6	14.8	18.2	17.3	22.2	134.2	4.9	136.2	8.0
C537	red beds	a	7/7	7/0	23.0	43.3	20.5	37.0	46.7	8.9	356.3	7.0
C538	red beds	a	5/9	2/3	4.6	9.5	4.1	27.5	199.7	5.4	189.3	18.0
C539	red beds	a	8/8	6/0	55.3	68.1	17.5	26.1	60.8	7.3	356.7	50.5
mean			17/19		28.7	67.9			3.1	24.5		
mean			17/19				12.4	33.5	35.0	6.1		
						Upper Jurassic						
C515	red beds	e	10/10	0/7	354.3	44.0	33.9	45.0	41.3	7.7	105.5	38.0
C516	red beds	e	6/6	0/2	358.8	49.6	18.9	38.9	131.2	6.6	71.6	22.8
C517	red beds	e	12/13	0/6	337.9	51.0	6.5	42.2	47.7	6.5	64.5	27.5
C518	red beds	e	8/8	8/0	331.6	43.8	22.2	45.3	84.7	6.1	88.5	47.0
C519	red beds	e	3/7	0/3	359.6	32.0	20.0	31.4	151.9	10.0	98.7	32.0
C520	volc. sed.	e	8/8	8/0	19.7	53.5	10.4	46.4	57.3	7.4	327.5	10.5
C521	volc. sed.	e	5/6	5/0	39.0	54.3	20.6	43.5	11.8	23.2	333.7	19.0
C522	volc. sed.	e	8/8	7/1	10.1	47.9	1.2	40.0	97.5	5.6	316.2	12.0
C523	red beds	e	4/4	0/4	49.9	55.6	15.5	58.0	46.2	13.7	296.7	22.0
C524	andesite	e	8/8	0/8	38.7	46.2	17.8	43.1	78.1	6.3	306.7	21.0
mean			10/13		8.3	50.5			19.6	11.2		
mean			10/13				16.7	43.7	74.0	5.7		
						Cretaceous						
C501	tuff	d	9/9	0/9	20.5	68.0	29.9	55.9	255.0	3.2	47.7	13
C502	tuff	d	8/8	0/8	5.2	56.8	15.1	40.9	100.5	5.6	38.7	18
C504	tuff	d	7/7	0/7	16.0	74.1	29.0	57.9	337.2	3.3	42.3	17
C505	tuff	d	8/8	0/8	4.2	74.8	18.4	56.7	1388.6	1.5	30.7	19
C525	red beds	e	9/9	9/0	14.0	57.5	20.2	49.4	269.1	3.1	46.3	9.3
C526	red beds	e	8/9	8/0	25.0	36.8	15.7	48.0	220.5	3.7	242.6	15.0
C527	red beds	e	9/9	4/0	343.3	52.7	6.7	44.1	207.1	3.8	62.9	22.0
C528	red beds	e	6/6	6/0	348.4	49.9	22.2	48.9	658.3	2.6	93.3	28.0
C529	red beds	e	6/9	6/0	21.8	30.2	16.7	50.4	425.9	3.3	215.7	21.0
C530	red beds	e	7/7	7/0	10.9	56.4	11.5	51.4	215.7	4.1	16.7	5.0
mean			10/11		9.4	56.6			23.7	10.1		
mean			10/11				18.0	50.6	131.5	4.2		

Abbreviations are Lith., lithology; volc., volcanic; sed., sediments; Loc., location on Figure 2; no/No, number of samples used to calculate mean direction/number of samples demagnetized; N/R, normal (N) or reversed (R) polarity (note that no polarity was assigned to samples fit with great circles); *D*, declination; *I*, inclination; *g*, in-situ (geographic) coordinates; *s*, tilt-corrected (stratigraphic) coordinates (at 100% unfolding); α_{95}, the radius that the mean direction lies within 95% confidence; *k*, the best estimate of the precision parameter; DD, dip direction of beds.

three-axis 2G cryogenic magnetometer. One site of Upper Jurassic andesites was demagnetized using an alternating field, while the rest of the samples were thermally demagnetized using an average of 16 temperature steps per sample. Results were analyzed using principal component analysis [*Kirschvink*, 1980] and *Fisher* [1953] statistics, or the *McFadden and McElhinny* [1988] great circle technique using *Fisher* [1953] statistics with a slight modification for the numbers of degrees of freedom. Maximum angular deviations of the least squares fits never exceed 20° and rarely exceed 10°. Reversal tests, performed on samples whose directions were determined by least squares fits, follow *McFadden and McElhinny* [1990]. Fold tests were performed in accord with *Watson and Enkin* [1993] using 1000 simulations and assuming a 5° circular standard deviation for bedding attitudes; uncertainties are reported at the 95% confidence level.

Middle Jurassic Zhuji and Zhougongshan Formations

Thermal demagnetization isolated two components of magnetization in most of the 156 samples collected: a north

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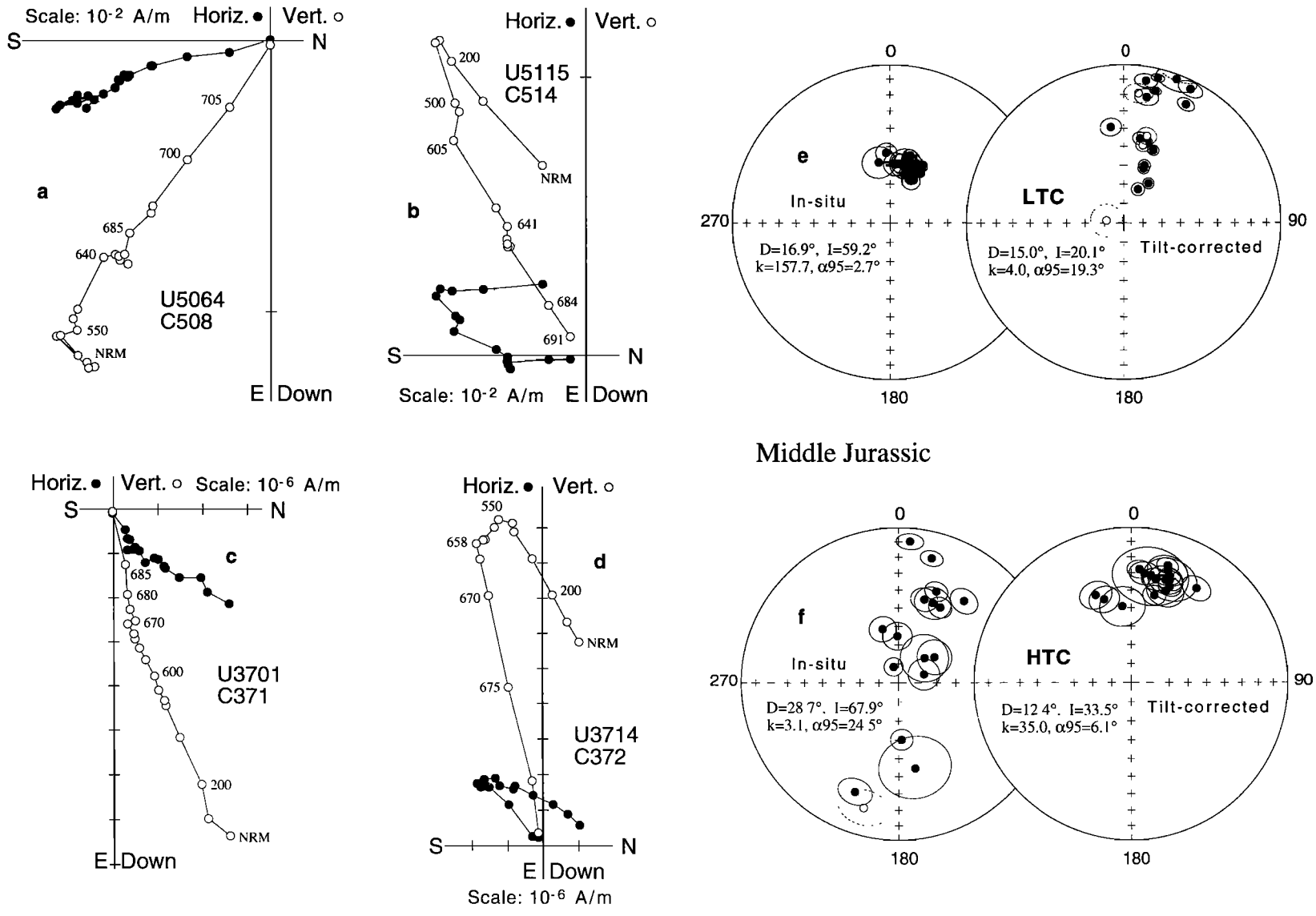


Figure 3. (a-d) Thermal demagnetization diagrams of Middle Jurassic red beds (in-situ coordinates). Temperature steps are in degrees Celsius; sites indicated on each plot correspond to Table 1. Stereonet plots of (e) low- and (f) high-temperature component locality mean directions; tilt-corrected directions shown at 100% unfolding. Site mean directions are plotted in the hemisphere best suited to demonstrate the clustering of the data.

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and down component (in-situ coordinates) at low temperatures (<550°C) and a north and down or south and up component (tilt-corrected coordinates) at high temperatures (>550°C) (Figures 3a to 3d). The low-temperature component (LTC), which normally does not trend toward the origin, was isolated in the samples from all 19 sites and defines a site mean direction of D=16.9°, I=59.2° (α_{95}=2.7°, k=157.7, N=19) whose concentration parameter is maximized at 1±2% unfolding (Figure 3e). This direction is significantly different from the expected geocentric axial dipole (GAD) (D=0°, I=51.3°) or present Earth's field (D=356°, I=47°) directions for the area and was unlikely acquired in a Brunhes field.

A high-temperature component that decays toward the origin was isolated in most of the samples from 17 of the 19 sites (Figures 3a to 3d). Two sites (C511 and C512) were omitted from further study because they possess a single direction of solely normal polarity that is indistinguishable from the LTC. Thirteen samples from the 17 sites were rejected from further study due to unstable (random) behavior during demagnetization, or because only the LTC could be isolated. For the remainder of the samples (127), principal component analysis was applied to all but 12 (9%), the latter of which were fit with great circles. The high unblocking temperatures (680°C) and the red color of the rocks suggest that hematite carries the remanence. For the high-temperature component (HTC), the Fisher precision maxima of the 17 site mean directions occurs at 107±6% unfolding, suggesting a prefolding remanence (Table 1 and Figure 3f). The reversal test is positive at 95% confidence limits. The presence of antipodal normal (N=75) and reversed (N=40) polarity, as well as the prefolding magnetization, support the interpretation that the HTC represents a primary magnetic remanence acquired in the Middle Jurassic.

We also performed anisotropy of magnetic susceptibility (AMS) measurements on 25 samples from sites C508, C514, C535, C537, and C539 (five samples per site) using a Kappabridge KLY-2 apparatus at the St. Maur Laboratory. The mean percentage of anisotropy (($k1/k3$ -1)x100) for each site is 1.0, 5.6, 26.8, 27.4, and 11.4, respectively, with the AMS ellipsoids being mostly oblate ($k2/k3 > k1/k2$) (Figure 4a) where $k1$, $k2$, and $k3$ are the normalized maximum, intermediate, and minimum principal eigenvalues, respectively. Because the $k3$ directions are oriented normal to bedding in tilt-corrected coordinates, a sedimentary origin for the AMS is suggested (Figure 4b). If hematite does contribute to the AMS signal, as found in other red beds [e.g., *Tauxe et al.*, 1990], the sedimentary fabric suggests that the magnetic remanence is detrital in origin [*Tauxe and Kent*, 1984]. The mean $k1$ direction trends in the same direction (D=283.3°, I=-5.4°) as the Middle Jurassic fold axes (Figure 2), indicative of a tectonic influence.

Upper Jurassic Maotanchang and Heishidu Formations

Of the 104 Upper Jurassic samples from 13 sites that were demagnetized, samples from 12 sites yielded a well-defined LTC whose direction trends north and down in in-situ coordinates and does not decay toward the origin (Figures 5a to 5d). For the 12 site means the concentration parameter maximizes at -11±3% unfolding. The prefolding magnetization direction (D=4.2°, I=52.3°, α_{95}=4.0°, k=117.7 at 0% unfolding) is virtually identical to the expected GAD direction and is probably a recent overprint (Figure 5e). The black shales (sites C531-C533) contain only the LTC. Approximately 90% of their magnetization is unblocked by 400°C; above that temperature, bulk susceptibility sharply increases and magnetization becomes unstable. They will not be discussed further.

The magnetizations of the red beds and red volcanoclastic sediments often decay toward the origin at high temperatures after removing the LTC (Figures 5a to 5c). Sometimes during demagnetization, the magnetization becomes unstable after trending away from the natural remanent magnetization direction; at other times the magnetization is entirely unstable (random) or only the LTC could be resolved. For the former case (25%), great circles were fit to those samples and then the great circle data was combined with best fit line data to calculate a site mean direction. The latter 6% were omitted from further analysis. As above, the red color and high unblocking temperatures indicate that hematite is the main carrier of the remanence. The andesite samples (C524) were demagnetized using alternating fields to 0.1 T which proved effective at isolating a characteristic direction (Figure 5d). Titanomagnetite likely carries their remanence.

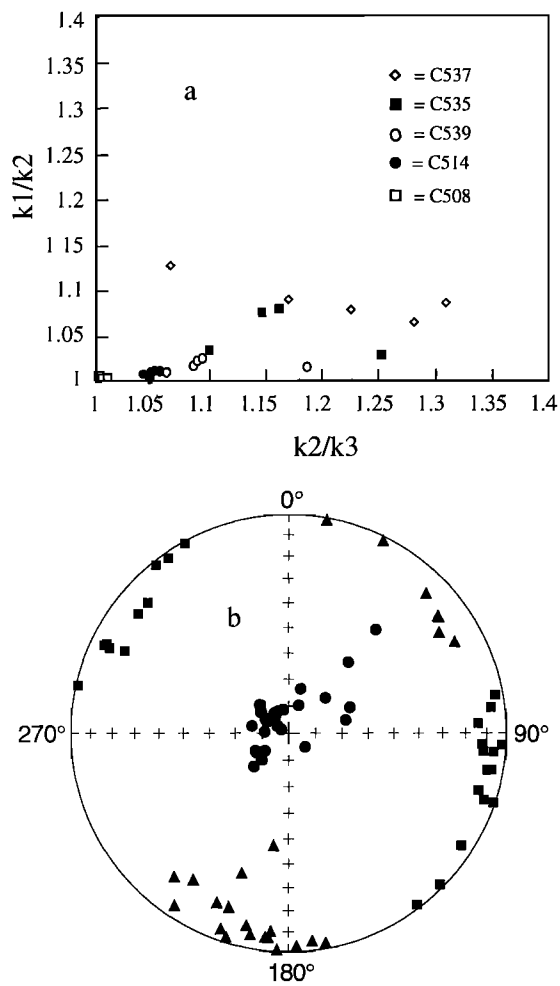


Figure 4. (a) Flinn diagram of anisotropy of magnetic susceptibility (AMS) data from five sites of Middle Jurassic red beds from this study. The normalized maximum, mean, and minimum principal values of the anisotropy eigenvalues are $k1$, $k2$, and $k3$, respectively. (b) Stereonet of the principal eigenvalue directions. Circles are $k3$, triangles $k2$, and squares $k1$.

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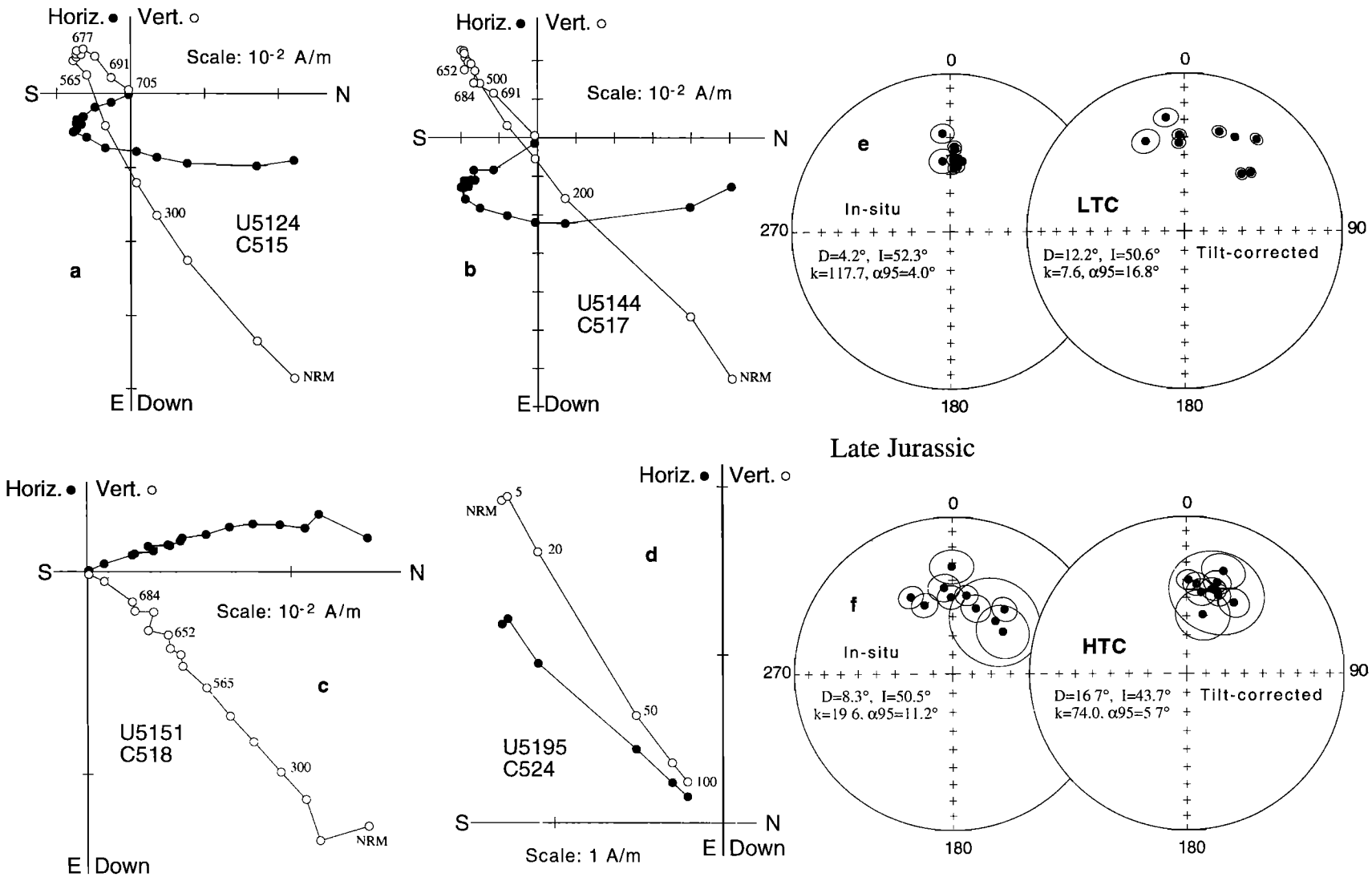


Figure 5. (a-c) Thermal demagnetization diagrams of Upper Jurassic red beds (in-situ coordinates). Temperature steps are in degrees Celsius; sites indicated on each plot correspond to Table 1. (d) Alternating field demagnetization diagram of Upper Jurassic andesite (in-situ coordinates). Cleaning steps are in milliteslas. Stereonet plots of (e) low- and (f) high-temperature component locality mean directions; tilt-corrected directions shown at 100% unfolding. Site mean directions are plotted in the hemisphere best suited to demonstrate the clustering of the data.

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The Late Jurassic HTC data are listed in Table 1. For the 10 site mean directions the concentration parameter is maximized at 78±14% unfolding. Although the 95% confidence envelope does not include the case at 100% unfolding, we do not consider that the remanence is of syn-folding origin for the following reasons: (1) the Late Jurassic HTC directions are of mixed polarity (normal=28, reversed=31), (2) the reversal test is positive, (3) the age of folding is post-Cretaceous, (4) the precision parameter maxima of the Cretaceous HTC direction includes the case at 100% unfolding (see below), (5) the direction at 78% unfolding (D=14.3°, I=45.5°) is indistinguishable at the 95% level of confidence from the 100% unfolding direction (Table 1), and (6) the fold test based on the F statistic [e.g., *McElhinny*, 1964] is positive at the 99% confidence level. Thus we consider that the HTC magnetization at 100% unfolding to be a primary, Late Jurassic remanence.

Cretaceous Chenpeng, Shibata, and Xiafuqiao Formations

The Cretaceous results come from two rock types: air fall tuffs and red beds. Thermal demagnetization of four sites of the red air fall tuff yields easily interpretable Zijderveld diagrams (Figures 6a and 6b). Two components are isolated: one that trends randomly but is often north and down directed (in-situ coordinates) at low temperatures (<300°C), and another with reversed polarity that decays toward the origin at high temperatures (>300°C). The red color and high unblocking temperatures (630°-680°C) suggests the magnetization resides in hematite. The purple-red colored ash fall tuff from site C503 yields highly dispersed directions. This site was rejected from further analysis.

The red bed sediments also yield easily interpretable demagnetization data. Two components were isolated, both of which are north and down directed (Figures 5c and 6d). The first component is isolated at low temperatures (<600°C) with a precision parameter maxima that occurs at 7±10% unfolding. The mean LTC direction of the red beds at 0% unfolding (D=8.5°, I=60.6°, α_{95}=3.0°, k=634.3) is similar to the mean postfolding Middle Jurassic LTC direction, and thus both mean LTC directions are likely post-Cretaceous in origin.

The four HTC site mean directions from the tuffs were combined with those from the six sites of red beds before calculating the overall site mean direction (Table 1). The Fisher precision maxima for the 10 sites occurs at 94±15% unfolding, suggesting that the remanence is prefolding (Figure 6f). A positive reversals test is achieved at the 95% confidence level. The prefolding, antipodal magnetization supports the interpretation that the HTC tilt-corrected direction represents a primary Cretaceous remanence.

Discussion

The mean tilt-corrected directions from Table 1 are shown in Figure 7. The data indicate a progression in direction from the Middle Jurassic to the Cretaceous as inclinations steepen and declinations rotate slightly clockwise. Table 2 lists the poles from this study together with other Mesozoic to Present poles from the NCB. Here the NCB is defined as the land south of the Yinshan-Tumen ophiolite [*Zhang et al.*, 1984] (i.e., south of the Mongolia Block in Figure 1), west of the Tan-Lu fault, and north of the Qinling suture. In our compilation of poles we follow *Zheng et al.* [1991] by omitting the Middle Jurassic result from *Fang et al.* [1988] because the details of sampling site, demagnetization methods, etc. are not provided. Furthermore, we do not accept the Middle Jurassic pole from the Yungan Formation [*Zheng et al.*, 1991] that is incorporated into other pole lists [e.g., *Enkin et al.*, 1992], because (1) there is no fold test available, (2) the directions are all normal (no reversals test), and 3) the mean in-situ (and tilt-corrected) direction (D=17.9°, I=58.7°, α_{95}=5.9°, N=3) is virtually identical to the mean Middle Jurassic and Cretaceous LTC directions determined in this study. We also reject the Middle to Late Jurassic studies of *Lin* [1984] from west of Tan-Lu (Mengying and Santai formations) as quoted in other pole lists [*Van der Voo*, 1993] because (1) no demagnetization data are presented, (2) every sample in the study contains normal polarity (no reversals test), and (3) no fold test is demonstrated. Thus the Late Jurassic pole from this study is the first of that age from the NCB to pass modern standards of reliability.

The other available Middle Jurassic pole from the NCB comes from the Ordos basin [*Yang et al.*, 1992]. The hypothesis that the Middle Jurassic pole from Ordos is indistinguishable from the coeval pole derived from sampling localities in Henan and Anhui provinces cannot be rejected at the 95% confidence level [*McFadden and Lowes*, 1981]. This is an important observation for two reasons: (1) because the strata from Ordos are flat lying, no possibility exists to perform a fold test, and thus the result from this study lends credibility to the data from Ordos, and (2) Ordos, the sampling localities from this study, and the region west of Ordos [*Wu et al.*, 1993] (Table 2) have acted as a rigid block, signifying these places have not been influenced by local vertical axis block rotations since the Cretaceous, and probably since the Middle Jurassic. Our pole position differs slightly from that of *Yang et al.* [1992] primarily because the inclination from our result is 8° shallower than that from Ordos. A principal difference between the two studies is that the carrier of the remanence from Ordos is primarily magnetite, while that from this study is hematite. Assuming the remanences from both studies are detrital, as suggested above for this study, it could be that either a systematic difference in grain shape, and/or different amounts of layer-parallel flattening accounts for the discrepancy in inclination.

Table 2 also lists Mesozoic to present poles from the SCB. The Mesozoic poles were derived from the compilations of *Enkin et al.* [1992] and *Gilder et al.* [1996]. We did not include the Cretaceous poles from places that were affected by Cenozoic rotations near the Red River fault (pole numbers 10 and 12 of *Gilder et al.* [1996]), nor the discordant pre-Cretaceous poles (numbers 24 and 36) obtained from localities southeast of the Shi-Hang Zone (Figure 1). The Shi-Hang zone probably demarcates displaced terranes to the southeast from more stable craton to the northwest [*Gilder et al.*, 1996]. The Middle Jurassic pole of *Zhu et al.* [1988] is not included in Table 2 because it is superseded by the result from *Huang and Opdyke* [1991].

The NCB-SCB Collision

Figure 8a shows a comparison of the apparent polar wander paths (APWP) for both the SCB and NCB, and Figure 8b shows the APWP for the NCB alone. Apparent in Figure 8a is that the poles are quite different before the Jurassic, then begin to

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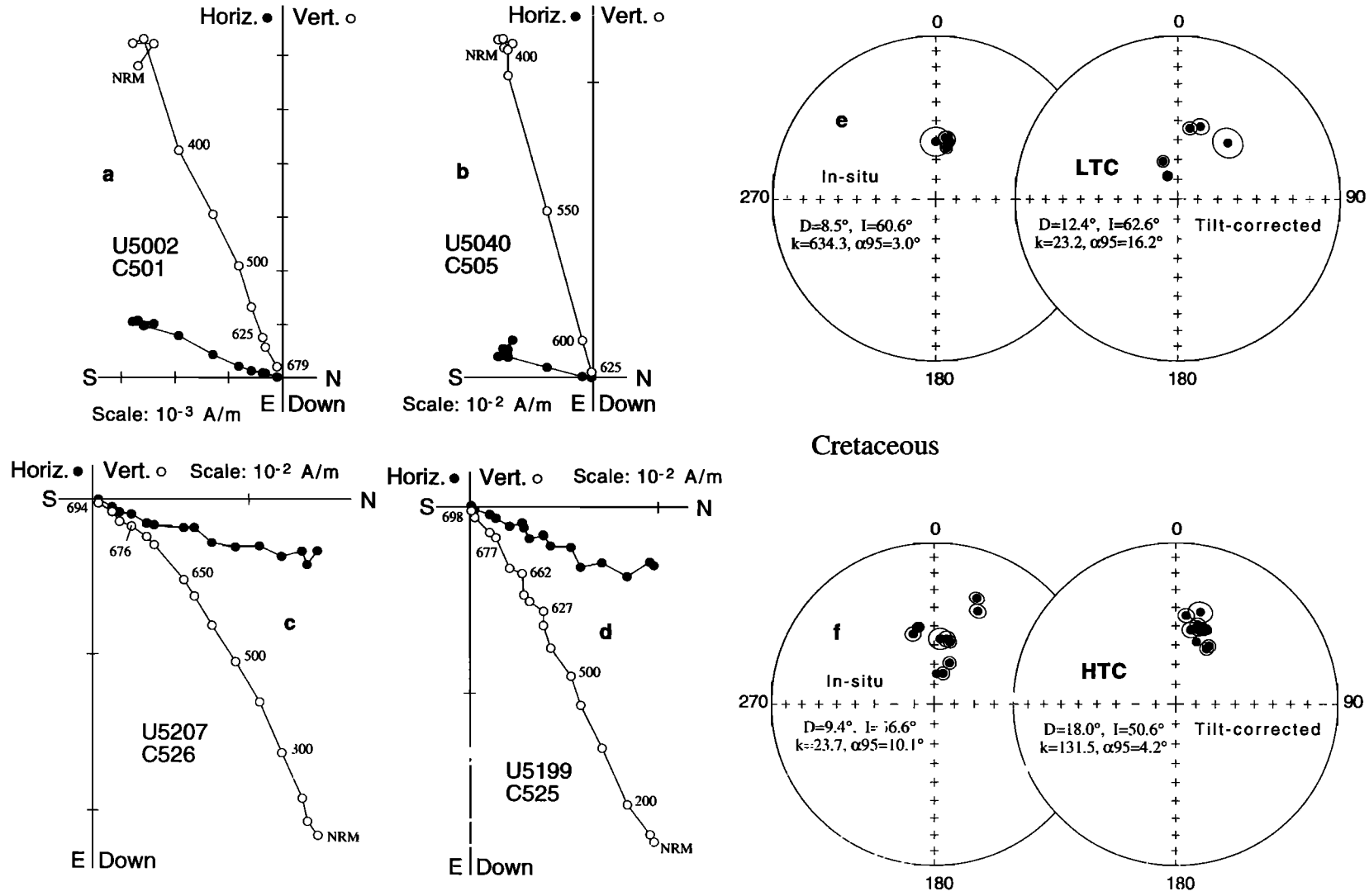


Figure 6. (a-b) Thermal demagnetization diagrams of Early Cretaceous air fall tuffs (in-situ coordinates). (c-d) Thermal demagnetization diagrams of Cretaceous red beds (in-situ coordinates). Temperature steps are in degrees Celsius; sites indicated on each plot correspond to Table 1. Stereonet plots of (e) low- and (f) high-temperature component locality mean directions; tilt-corrected directions shown at 100% unfolding. Site mean directions are plotted in the hemisphere best suited to demonstrate the clustering of the data.

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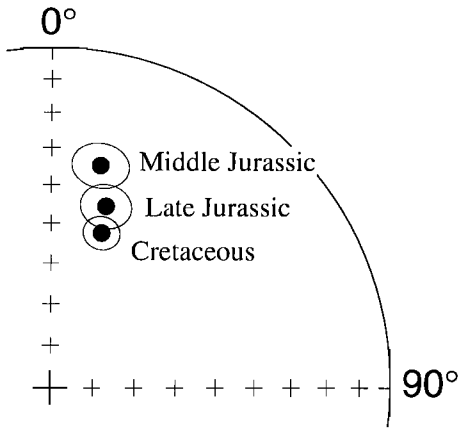


Figure 7. Comparison of the mean Middle Jurassic, Late Jurassic, and Cretaceous tilt-corrected directions (100% unfolding) determined in this study.

coalesce The mean Late Jurassic poles from both plates are statistically identical, suggesting that the collision was complete by that time, as proposed by *Huang and Opdyke* [1991]. A more precise estimate for the end of collision is hindered by the error and wider age window of the Early to Middle Jurassic pole from the SCB. However, a comparison of that pole with the Middle Jurassic pole from the NCB indicates that the two do not share a common mean at the 95% significance level [*McFadden and Lowes*, 1981].

From the APWPs we calculated the declinations and paleolatitudes for a reference point at 31.6°N, 116°E (near the center of our study area) and plotted them as a function of time in Figures 8c and 8d It is clear that the relative differences in declination account for the discordance of the pole paths. This implies that suturing progressed as the two plates rotated relative to each other, consistent with the scissor model of the collision proposed by *Zhao and Coe* [1987, 1989]. The small difference in paleolatitude and large difference in declination calculated for a point near the suture zone also implies that the reference point (31.6°N, 116°E) must lie near (<500 km) the pole of rotation of the two plates.

Some of the features in Figures 8c and 8d require further comment For example, there appears to be significant deviations in paleolatitude around the Early to Middle Jurassic, whereas little to none occur before or after those times. However, it is unlikely that the NCB was south of the SCB at that time and it seems that the paleolatitude for the NCB is too shallow, while that for the SCB is too steep, when compared to the overall trend of the curves. This discrepancy probably results from the limited number of Early to Middle Jurassic data. Inclination shallowing should not greatly influence the interpretation of the paleolatitude data because the lithology and magnetic mineralogy from both plates are largely the same; if inclination shallowing is a factor, both curves should be equally affected

Another way to estimate the final age of collision from paleomagnetic data is based on the track and cusp (hairpin turn) concept [e.g., *Francheteau and Sclater*, 1969; *Irving and Park*, 1972]. According to this model, plate motions are assumed to be uniform over large time intervals. Translated into paleomagnetic terms, this means that the poles from a plate will lie along small circle swaths, called tracks, that are centered on paleomagnetic Euler poles [*Gordon et al.*, 1984]. When boundary conditions change in a major way, such as the closure of an ocean basin between two plates due to full collision, plate boundary forces are significantly altered, plate motions are reorganized, and the trajectory of the new plate defines a new track. The intersection of two tracks is called a cusp, which is believed to correspond to the time of plate reorganization.

It appears that the Mesozoic APWP from the NCB contains two tracks (Figure 8b): one from the Early Triassic to the Middle or Late Jurassic and another from the Middle or Late Jurassic through Cretaceous. The timing of the change in track path coincides well with the change in orientation from the east-west trending (suture parallel) axial planes of the Middle Jurassic red beds to the north-south trending axial planes of Upper Jurassic and younger rocks. Because (1) fold axis trends changed between the Middle and Late Jurassic, (2) the mean Middle Jurassic pole from the NCB is discordant from the SCB Early to Middle Jurassic pole, and (3) major plate reorganization of the plate trajectories (cusp) occurred during the Middle to Late Jurassic, we conclude that the best estimate for the end of collision and associated deformation is near the Middle to Late Jurassic boundary; e.g., 159+10/-8 Ma [*Gradstein et al.*, 1994]. The youngest U-Pb date of an eclogite-facies metamorphic rock in the suture (135±42) is consistent (within error) with this timing [*Ames et al.*, 1996].

If collision, as defined by first contact of plates, began in the latest Paleozoic [*Laveine et al.*, 1987; *Zhao and Coe*, 1987], then the collision lasted about 100 million years (m.y.) from beginning to end. Our estimate for the end of collision is younger by approximately 55 m.y. than the peak in Ar-Ar and U-Pb dates of the high-pressure facies metamorphic rocks found in the suture (Figure 8e), implying that ages of metamorphic rocks in suture zones represent a more extreme phase of the collision It is interesting to note that the bulk of these dates, between 225 and 190 Ma, correspond to the period of fast relative angular velocity between the SCB and NCB (roughly 1°/Ma, Figures 8d and 8e). It has yet to be explained why a pulse of igneous activity occurred along the suture at about 125 Ma

Collision Between the United China Block and Eurasia

We argue above that the collision between the NCB and SCB was complete by the middle Late Jurassic. Assuming this is true, we can average the Late Jurassic and Cretaceous poles from the two blocks before comparing them with the Eurasian reference path [*Besse and Courtillot*, 1991]. We further assume that the collision between the united China block (UCB) and Mongolia was completed by the Late Jurassic based on the proximity of Permian poles from Mongolia (MON) and the NCB [e.g., *Zhao et al.*, 1990] and average the Mesozoic poles from MON together with those from the UCB. Only one Mesozoic study from Mongolia demonstrates a positive field test; namely, that for the Late Jurassic reported by *Zhao et al.* [1990] (68.5°N, 231.6°E, A_{95}=9.5°, N=4). Averaging this result with the four studies from Table 2 yields the mean Late Jurassic pole for UCB+MON which is 73.7°N, 218.8°E (A_{95}=5.9°, N=5 studies). For the UCB+MON plate, the mean Cretaceous pole lies at 74.7°N, 212.5°E (A_{95}=4.2°, N=15 studies) We averaged the Early and Late Cretaceous poles together because both populations are statistically identical at

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Table 2. Mesozoic Paleomagnetic Poles From the NCB and SCB

Age	Lat.	Lon.	λ_p	ϕ_p	A_{95}	No	Reference
				North China Block			
Ng_2	36.2	118.5	76.4	177.2	7.7	4L	*Lin et al.* [1985]
Ng_2	41.0	114.7	83.4	192.8	6.2	9L	*Zheng et al.* [1991]
Ng_2	36.2	118.5	84.3	233.5	6.8	9L	*Zhao et al.* [1994]
Ng_2	36.4	118.8	87.2	272.4	16.3	3L	*Zhao et al.* [1994]
Ng_2 Mean	-	-	83.8	199.4	7.1	4S	
K_1	42.0	119.2	84.9	258.8		4L	*Zhao et al.* [1990]
K_2	40.1	112.9	79.6	170.1	5.8	4L	*Zheng et al.* [1991]
K_{1-2}	37.2	105.0	74.5	203.4	8.2	10L	*Wu et al.* [1993]
K_1	35.0	108.0	75.8	208.7	7.5	10L	*Ma et al.* [1993]
K_{1-2}	31.6	116.0	74.5	201.0	4.7	10L	this study
K_{1-2} Mean	-	-	78.6	202.6	6.2	5S	
J_3 Mean	31.6	116.0	74.4	222.8	5.9	10L	this study
J_2	36.2	109.3	74.3	232.8	5.0	6L	*Yang et al.* [1992]
J_2	31.6	116.0	72.9	254.7	6.4	17L	this study
J_2 Mean	-	-	73.6	249.3	4.9	23L	
J_1 Mean	36.2	109.3	82.4	286.0	10.0	10L	*Yang et al.* [1992]
T_{2-3} Mean	-	-	60.2	9.6	11.6	4S	*Enkin et al.* [1992]
T_1 Mean	-	-	53.4	358.4	8.9	5S	*Enkin et al.* [1992]
				South China Block			
Pg_1 Mean	-	-	85.4	241.7	13.5	3S	*Zhao et al.* [1994]
K_{1-2} Mean	-	-	72.6	217.7	5.5	10S	*Gilder et al.* [1996]
J_3	-	-	73.0	213.7	12.6	6L	*Lin et al.* [1985]
J_3	29.5	103.7	74.6	235.3	13.2	5L	*Enkin et al.* [1991]
J_3	26.7	102.4	74.4	185.1	3.9	20L	*Huang and Opdyke* [1991, 1993]
J_3 Mean	-	-	74.9	211.4	10.3	3S	
J_2	26.6	106.7	67.8	185.3	12.0	18s	*Lin* [1984]
J_1	26.8	102.3	63.8	188.2	16.1	8L	*Huang and Opdyke* [1991]
J_1	22.6	108.5	60.4	204.5	14.0	3L	*Gilder et al.* [1993] (Area 2- adj.)
J_{1-2} Mean	-	-	64.3	193.5	9.0	3S	
T_{1-2} Mean	-	-	43.0	214.2	7.7	7S	see text
P_2-T_1 Mean	-	-	43.4	218.4	8.1	4S	see text

Abbreviations are Lat., degrees north latitude of sampling locality, Lon., degrees east longitude of sampling locality, Ng, Neogene; Pg, Paleogene; K, Cretaceous; J, Jurassic; T, Triassic; P, Permian; 1, Early; 2, Middle; 3, Late (2, Late for the Permian and Cretaceous); λ_p, degrees north latitude of pole; ϕ_p, degrees east longitude of pole; A_{95}, the angular radius of the cone within which the mean pole lies within 95% confidence; No, number of localities (L), samples (s), or Studies (S) used to determine pole.

95% confidence limits, consistent with the hypothesis that a standstill occurred during the hairpin turn on the Eurasian APWP between 130 to 70 Ma [*Besse and Courtillot,* 1991].

We plotted these poles together with the Eurasian reference APWP in Figure 9. It is apparent in Figure 9 that the Late Jurassic UCB+MON pole lies far from that for Eurasia (70.3°N, 161.5°E, A_{95}=5.4°, N=3), requiring 17.3±6.2° (about 2000 km) of north-south directed shortening with negligible relative rotation. A comparison of the Cretaceous poles as seen from 31.6°N, 116°E suggests that neither significant shortening (2.8±3.5°) nor rotation (0.7±4.0°) are required to resolve them. Not only is the end of collision well defined by the concordance of the Cretaceous poles but, like the case for the NCB-SCB collision, the end of suturing is marked by a cusp on the Eurasian APWP. Although it has been previously recognized that the collision coincides with the Eurasian Cretaceous hairpin turn [e.g., *Enkin et al.,* 1991], the small uncertainties in our compilation make this a robust

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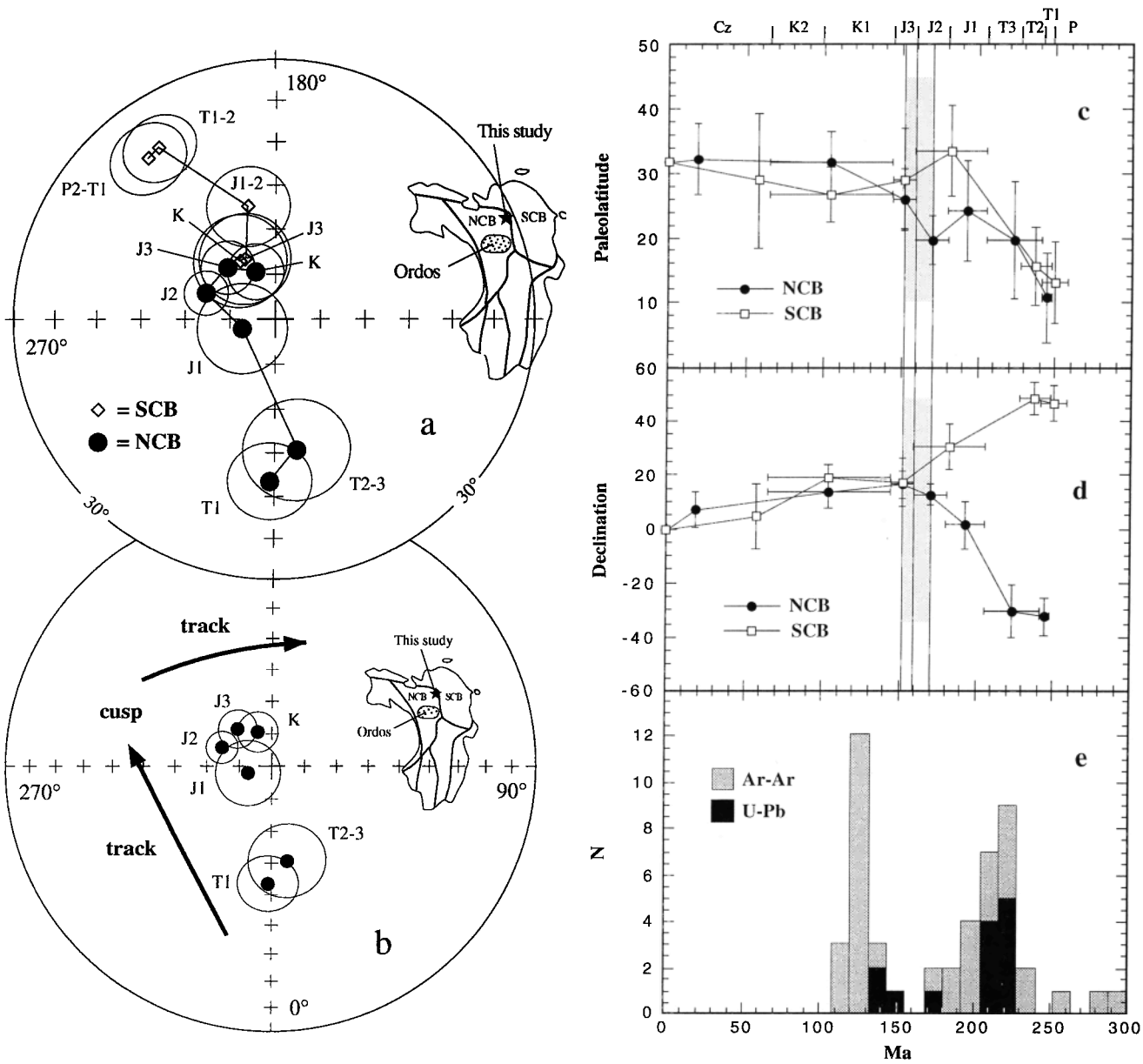


Figure 8. (a) Comparison of SCB (diamonds) and NCB (circles) Mesozoic APWPs (data from Table 2). (b) The APWP of the NCB, contains two tracks separated by a cusp at about the Middle to Late Jurassic. (c) Paleolatitude and (d) declination plotted as a function of time. Values calculated for a point near the center of our study area at 31.6°N, 116°E; shaded area represents 159+10/-8 Ma. (e) Histogram of post 300 Ma U-Pb and Ar-Ar dates of granites and metamorphic rocks from the Qinling and Sulu suture zone (data from *Reischmann et al.* [1990], *Mattauer et al.* [1991], *Ames et al.* [1993, 1996], *Eide et al.* [1994], and *Hacker and Wang* [1995]). Note that (1) the peak around 215 Ma corresponds to the time of rapid relative rotation between the NCB and SCB, (2) the peak at 125 Ma is related to granite intrusion and volcanism found along the length of the suture, and (3) the youngest U-Pb zircon date from eclogite is 135±42 Ma [*Ames et al.*, 1996]. Abbreviations above figure are Cz, Cenozoic; K, Cretaceous; J, Jurassic; T, Triassic; P, Permian; 1, Early; 2, Middle; 3, Late (2, Late for the Cretaceous).

conclusion. One problem that remains, however, is identifying the geological manifestation of the collision. Although the Mongol-Okhotsk suture zone is well defined by ophiolites and high-pressure facies metamorphic rocks in its eastern part [*Zonenshain et al.*, 1990] (Figure 1), where the suture continues to the west is as yet unknown.

Another cusp on the Eurasian path occurs at 50 Ma (Figure 9). This cusp has been attributed to the India-Asia collision [e.g., *Besse and Courtillot*, 1991]. The expected direction at the site localities of this study from the 50 Ma Eurasian path is D=8.8°, I=60.5°, which is very similar to the mean Middle Jurassic and Cretaceous LTC directions discussed above yet unlike the Late Jurassic LTC direction which resembles a Brunhes overprint. A widespread remagnetization occurring at about 50 Ma along the Mongol-Okhotsk suture in Siberia was found by *Halim et al.* [1996]. Tectonic reactivation of both

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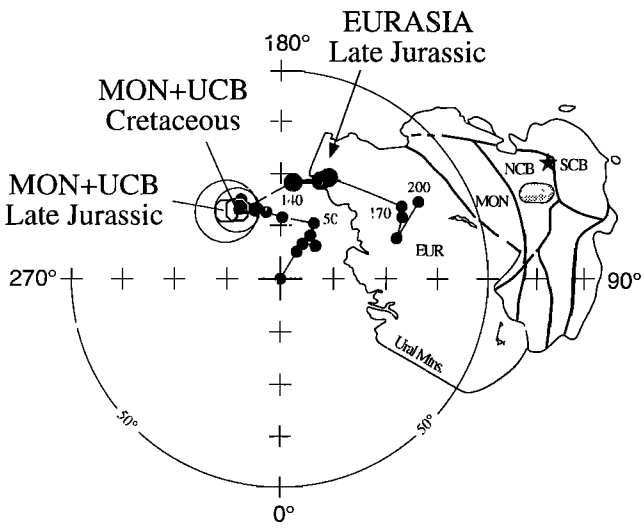


Figure 9. Late Jurassic and Cretaceous poles from MON+UCB (open symbols) compared with the Eurasian reference path (closed symbols, each symbol = 10 m.y.) [*Besse and Courtillot,* 1991]. Uncertainties in the individual poles from Eurasia are generally between 3° and 5°. Dates next to the Eurasian path are in mega annum; the Late Jurassic segment of the Eurasian path is represented by a thicker line and larger symbols. Note that a standstill occurs in the Eurasian APWP between 70 and 130 Ma [*Besse and Courtillot,* 1991].

the Mongol-Okhotsk and Qinling suture zones due to the India-Asia collision has been proposed, and it is possible that the non-Brunhes LTC partial remagnetization is related to the latest continental collision in Eurasia.

Conclusions

Pre-Late Jurassic poles from the NCB and SCB are significantly different at the 95% confidence level. Their Late Jurassic poles are indistinguishable, implying that the collision of the two plates was over before or during the Late Jurassic. Near Middle Jurassic time, the uncertainties of the paleomagnetic poles and larger time window of the pole from the SCB hampers a more precise answer as to when collision terminated. Middle Jurassic fold axes and AMS maximum eigenvalue directions are oriented parallel to the suture, and some Middle Jurassic beds are overturned, suggesting that compression normal to the suture continued after the Middle Jurassic. Upper Jurassic and Cretaceous fold axes are oriented orthogonal to the suture, and the bedding dips of these rocks rarely exceed 30°, implying that compression acting perpendicular to the suture ceased in the Late Jurassic. These observations support published stratigraphic sections that show an angular unconformity separating Middle Jurassic from younger beds. Taken together, the best estimate for the end of collision is near the Middle to Late Jurassic boundary with an uncertainty that extends to the means of those series, namely, 159+10/-8 Ma [*Gradstein et al.*, 1994].

For the case of the NCB-SCB collision, paleomagnetic studies have successfully identified the timing and kinematics of the entire collision process, suggesting that the two plates first contacted in the late Paleozoic and were fully sutured by about 160 Ma. Paleontologic data coincide well with the paleomagnetic data regarding when the plates came into contact. The scissor-like closure was accompanied by a suture-parallel change from marine to continental sedimentation, with the transition younging westward. The time of rapid relative rotation (1°/Ma) is marked by a peak in U-Pb and Ar-Ar dates of rocks associated with the collision. The end of collision is characterized by a regional angular unconformity, a change in the major strain axis orientation from suture-normal to suture-parallel, concordance of the paleomagnetic poles from both plates, and also a cusp on the NCB APWP. Indeed, the identification of cusps in APWPs may be another way to date continental collisions. The cusp near the Middle to Late Jurassic part of the NCB APWP occurs when the NCB and SCB poles converge. This cusp would thus carry the same kind of memory of a major collisional event, as do the 130-70 Ma and 50 Ma features which shape the Eurasian APWP.

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